

Cloud LearnX

Mathematics

Transformations — Practice Worksheet

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Topic: Transformations

Instructions:

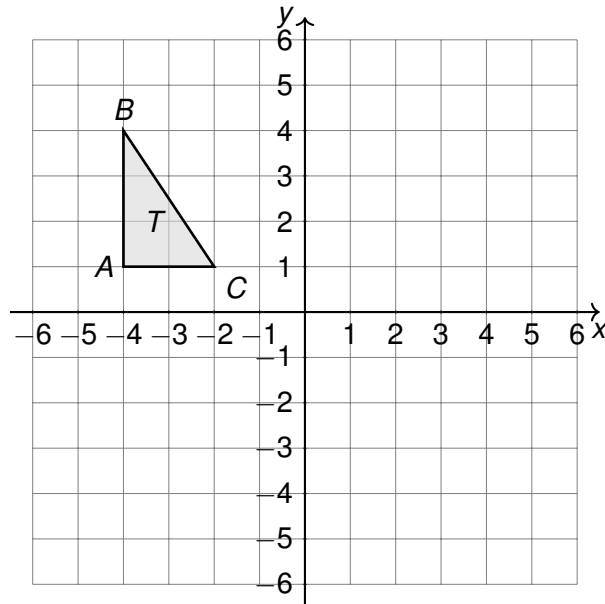
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 Triangle T is drawn on the grid below.



Translate triangle T by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$.

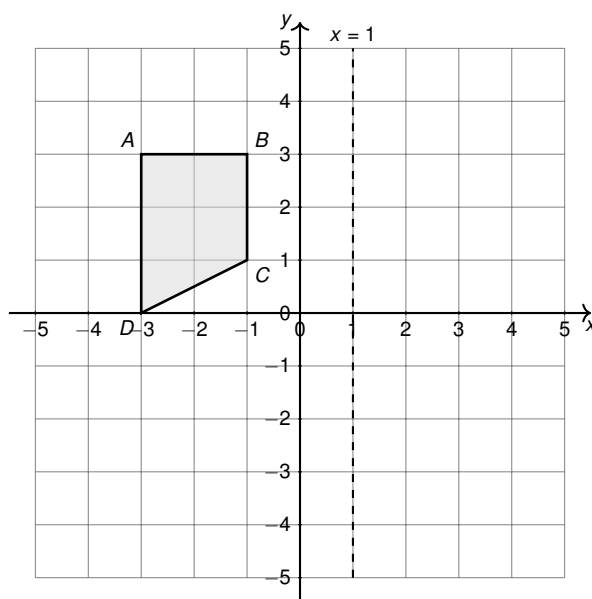
Label the image T' .

(2)

(Total for Question 1 is 2 marks)

2 Quadrilateral Q has vertices $A(-3, 3)$, $B(-1, 3)$, $C(-1, 1)$ and $D(-3, 0)$.

The line $x = 1$ is drawn on the grid below.



Reflect quadrilateral Q in the line $x = 1$.

Write down the coordinates of the image of each vertex.

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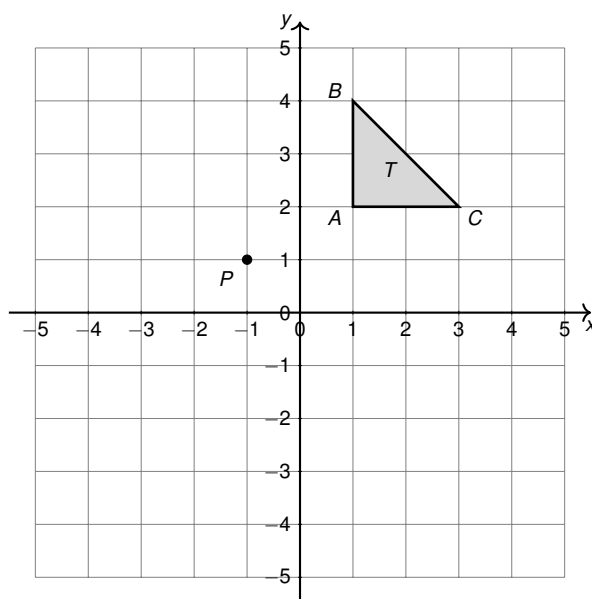
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(2)

(Total for Question 2 is 2 marks)

3 Triangle T has vertices $A(1, 2)$, $B(1, 4)$ and $C(3, 2)$.

The point $P(-1, 1)$ is marked as the centre of rotation.

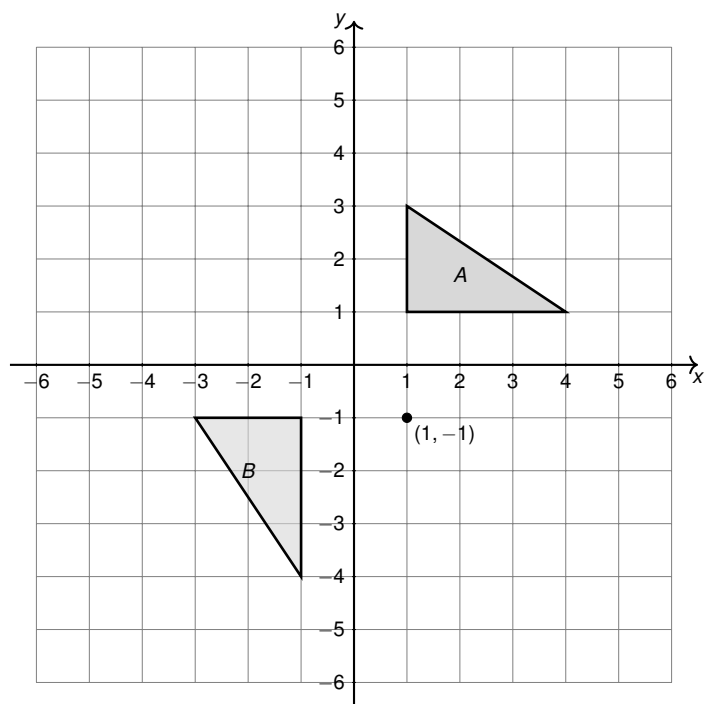


On the grid, draw the image of triangle T after a rotation of 90° clockwise about the point P .

(3)

(Total for Question 3 is 3 marks)

4 Shape A and shape B are drawn on the grid below.



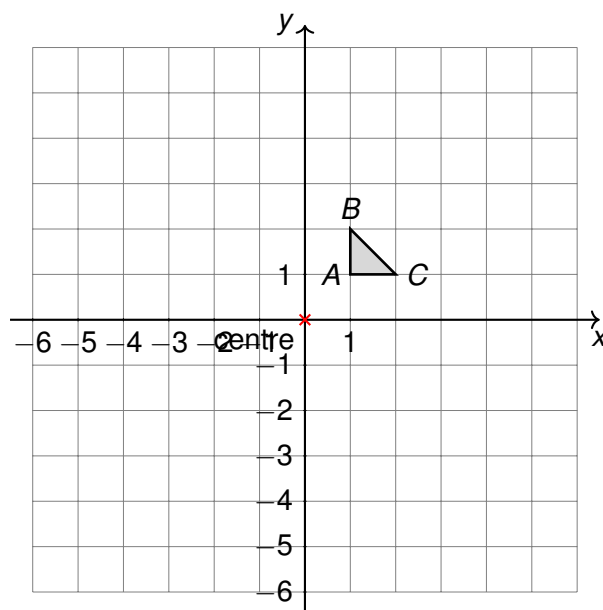
Describe fully the single transformation that maps shape A onto shape B .

(3)

(Total for Question 4 is 3 marks)

5 (a) Triangle P has vertices $A(1, 1)$, $B(1, 2)$ and $C(2, 1)$.

Enlarge triangle P by scale factor 2, centre $(0, 0)$.

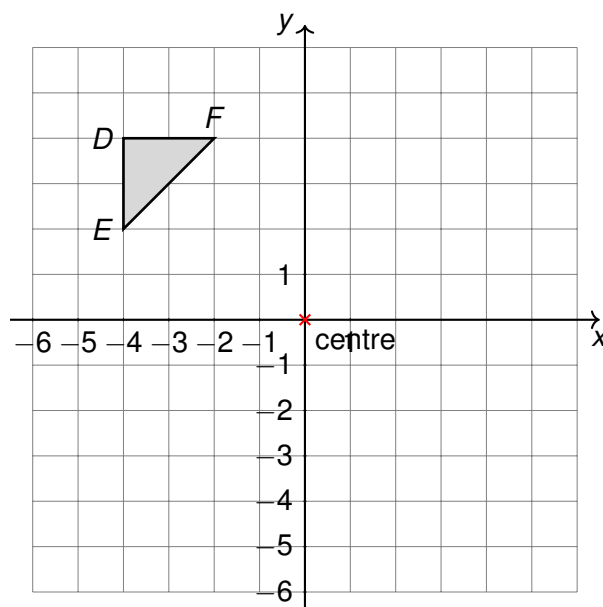


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(2)

(b) Triangle Q has vertices $D(-4, 4)$, $E(-4, 2)$ and $F(-2, 4)$.

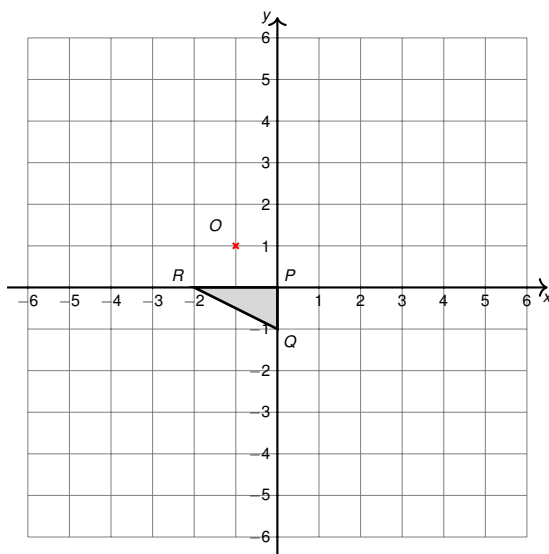
Enlarge triangle Q by scale factor $\frac{1}{2}$, centre $(0, 0)$.



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(2)

(Total for Question 5 is 4 marks)

- 6 In a school science lab, a microscope projects a slide image onto a screen. The lens system inverts the image and enlarges it by scale factor -2 about a fixed point O on the screen (the optical centre). Triangle PQR represents a specimen shown on the slide, with O marking the optical centre of the lens. The coordinates are shown on the grid below.



- (a) On the grid, draw the image of triangle PQR after an enlargement of scale factor -2 , centre $O(-1, 1)$. Label the image $P'Q'R'$.

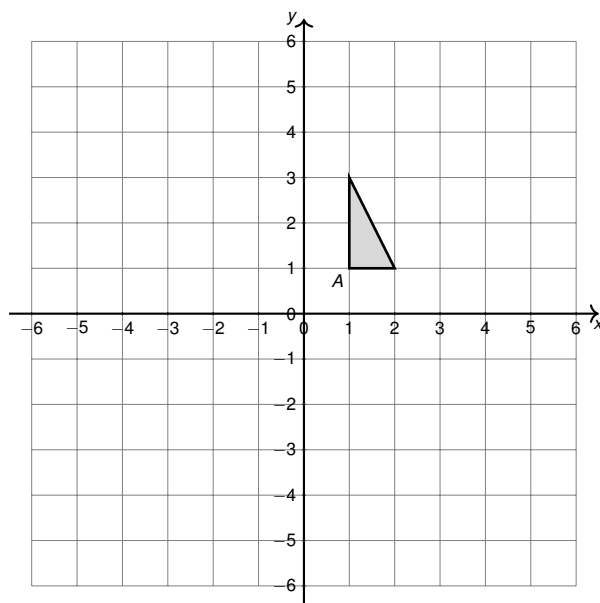
- (b) State the coordinates of P' , Q' and R' .

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(4)

(Total for Question 6 is 4 marks)

7 Triangle A has vertices $(1, 1)$, $(1, 3)$ and $(2, 1)$, as shown on the grid below.



- (a) Triangle A is translated by the vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ to give triangle B .

Triangle B is then reflected in the y -axis to give triangle C .

Find the coordinates of the vertices of triangle C , and draw triangle C on the grid above.

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(2)

- (b) Show that the single transformation that maps triangle A directly onto triangle C cannot be a translation.

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(2)

(Total for Question 7 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS